

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Built a hands-off attendance-enforcement pipeline that runs a daily in-database cron job, resolves absentees, and posts targeted Slack reminders, automating a three-strikes rule that previously required manual tracking.
- Built the club's mentorship lineage as an interactive 3D radial graph in react-three-fiber, hand-writing a proportional-arc layout that prevents edge crossings and a custom camera that autofits each family to the viewport.

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Designed a dual-database layer that keeps transactional records in PostgreSQL while serving relationship-heavy reads from Neo4j, routing each query class to the store that answers it fastest instead of forcing one engine to do both jobs.
- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Owned the Spring Boot backend serving the platform's core CRUD and graph-traversal APIs, structuring endpoints around the dual-store model so the client never has to know which database backs a given read.

PROJECTS

fliks | *Go, PostgreSQL, AWS EC2*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Architected an ML rating service that grades each clip independently and shows its verdict beside the crowd's, with those crowdsourced ratings serving as the labeled data the classifier trains on.
- Coordinated transcode jobs through a Postgres-backed queue using SELECT FOR UPDATE SKIP LOCKED instead of bolting on Redis or SQS, keeping the whole system on one datastore and one self-provisioned EC2 box.
- Made ratings immutable and gated commenting behind a submitted rating, so the certification signal cannot be retroactively gamed and social participation feeds skill data instead of diluting it.

Dadei | *Python, pgvector*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Replaced single-signal context lookup with hybrid retrieval over pgvector that fuses vector similarity, lexical overlap, recency decay, and participant overlap, grounding each proposal under a fixed token budget.
- Engineered a no-merge-first speaker-ID model using an EMA voice centroid plus a bounded prototype bank with floor-and-margin gating, abstaining into a new identity on ambiguity, attributing 96% of utterances to the correct speaker.

WizViz | *Python, MediaPipe, OpenCV*

devpost.com/software/wizviz

Webcam fighting game driven by real-time body-gesture recognition, winner of the Interactive Media track at SpartaHack X

- Ran pose inference asynchronously off the render loop with a monotonic timestamp guard that drops out-of-order results, sustaining 60 FPS with sub-20ms input lag instead of blocking on per-frame detection.
- Classified five distinct combat gestures from body landmarks with a scale-invariant geometric model normalized by torso length, making detection work across players and camera distances without any labeled training data.

TECHNICAL SKILLS

Languages Python, SQL, Java, JavaScript/TypeScript

Databases PostgreSQL, Neo4j, Redis, pgvector

Cloud & Infrastructure AWS (EC2, S3, RDS)